

Bibliographic Review, Topic 39

The Blind Spot Paradox

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1 Defining and classifying drift

The vocabulary was unstable for years, and Moreno-Torres et al. [1] write their taxonomy to fix it: covariate shift when only $P(X)$ moves, prior probability shift when $P(y)$ does, and concept shift, the case usually called concept drift, when $P(y | X)$ changes. The contribution is terminological rather than algorithmic, and they say so.

Two surveys map the field. Gama et al. [2] organise adaptation into forgetting mechanisms, ensembles and explicit detectors; Lu et al. [3] restructure the same material around detection, understanding and adaptation, and classify drift by shape. Both work from one assumption: the detector serves the classifier, the classifier reacts to the detector, and neither framework anticipates the two working against each other.

2 Adaptive learners on streams

Domingos and Hulten [4] introduced VFDT, a decision tree that reads each example once in constant time and memory, using a Hoeffding bound to decide when a node has seen enough data to split with confidence. It agrees asymptotically with a batch learner on stationary data, and it never undoes a split, so it does not adapt.

ADWIN [5] adds the adaptivity VFDT lacks. It keeps a window of variable length, cuts it at every possible point, and declares a change when two sub-window means differ by more than a Hoeffding bound. Unlike most competing heuristics it comes with proven bounds on false positive and false negative rates, and its second version runs in $O(\log W)$ memory and update time.

The Adaptive Random Forest [6] combines the two: M Hoeffding trees, Poisson($\lambda=6$) resampling as in leveraging bagging, and per-tree detection with two thresholds, a permissive δ_w that starts a background tree and a stricter δ_d that promotes it. ARF is evaluated on predictive accuracy alone, as is conventional for ensembles of this kind. River [7] is the Python library implementing these algorithms, and its defaults are what deployed pipelines inherit.

3 Detecting change from an error stream

Page [8] proposed the cumulative sum for industrial quality control, against moving averages whose window length forces a choice between reacting to large shifts and detecting small ones. CUSUM accumulates deviations from a reference level and fires when the running sum passes a threshold, so its delay grows as that threshold divided by the size of the shift. Hinkley [9] takes the same statistic and asks where the change occurred rather than whether it did, deriving the asymptotic law of that estimate for normal observations with known initial mean and variance. Both assume a change that persists.

KSWIN [10] belongs to another family, comparing two sliding windows with a non-parametric Kolmogorov-Smirnov test and pairing detection with prototype-based adaptation. Its authors report more false positives than ADWIN or DDM, and argue they are harmless because the adaptation they trigger is cheap.

4 Evaluation and synthetic ground truth

Gama et al. [2] describe three protocols: holdout, which needs a test set drawn from the concept active at time t and is therefore restricted to synthetic data; prequential, where each instance is tested before training, over a landmark window, a sliding window or fading factors; and controlled permutations, which rerun the stream in randomised but locality-preserving orders. Lu et al. [3] add the metrics: kappa and kappa-temporal for imbalanced or temporally dependent labels, prequential AUC, RAM-hours for cost. Neither lists a measure of how long a model takes to recover.

Ground truth is the obstacle in finance, since nobody knows when a real regime actually changed. ProteuS [11] fits ARMA-GARCH models to four ETFs and simulates abrupt and gradual transitions between the fitted regimes with known break dates. Its authors' own analysis finds heavy overlap between the market states, so the task stays hard.

5 The source paper

The Blind Spot Paradox [12] attacks the cooperative assumption. Instrumenting the ARF's swap counter in River 0.23.0, it argues that an adaptive classifier can erase the evidence an external detector needs. The first mechanism is a clock artefact: River evaluates ADWIN every c steps, 32 by default, while the ARF's internal detectors run at $c = 1$, which handicaps the external monitor by construction. The second, named the Hydra effect, is that the forest adapts as soon as its fastest tree does, measured 4 to 8 times faster than a single tree. The third is starvation:

the transient error excess and the CUSUM accumulation time both scale as $O(1/\Delta e)$, so a violent shock is repaired fast enough to leave no more evidence than a slow one. On heteroscedastic ProteuS streams a Page-Hinkley test paired with an ARF at $c_{\text{int}}=1$ detects none of 1,080 runs. The proposed Decoupling Principle asks that the external monitor be strictly more reactive than the internal one, and the paper reports that on those streams this conflicts with false-alarm control, leaving no admissible threshold. A windowed KSWIN monitor escapes the effect altogether.

The paper is explicit about what it leaves unsettled. The starvation boundary assumes conditional independence between trees, which the shared stream violates, and the authors note this is why the measured acceleration falls short of the M -fold reference. Swap tracking mixes genuine detections with background noise, which rules out isolated ARL_0 comparisons. And the predicted starvation appears only on stationary synthetic streams: on real non-stationary data the failure inverts into a collapse of precision through false-alarm flooding.

6 Statistical background

For the law of $\min_i \tau_i$ the paper draws on classical order statistics [13], whose standard result gives the expected minimum of M independent variables as $\int_0^\infty (1-F)^M$. That says nothing once the variables are dependent. Esary, Proschan and Walkup [14] handle that case, defining association through $\text{Cov}[f(T), g(T)] \geq 0$ for all non-decreasing f and g , and proving that associated variables satisfy $\Pr(\tau_1 > s_1, \dots, \tau_M > s_M) \geq \prod_i \Pr(\tau_i > s_i)$, the reverse of the independent case.

A separate literature deals with two competing stopping times. Fine and Gray [15] model the cumulative incidence function $F_1(t) = \Pr(T \leq t, \varepsilon = 1)$, the probability of failing from a given cause by t , arguing that the cause-specific hazard has no direct reading in those terms; their own caveat is that the corresponding risk set is artificial, since it keeps subjects who have already failed from another cause. Aalen and Johansen [16] give the estimator this needs, a product-limit estimator of the transition probabilities of a non-homogeneous Markov chain under censoring.

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